

Threshold Tomography for Latent Curve Recovery

A Tomographic Inverse Problem with Binary (Thresholded) Measurements

This is a **tomographic inverse problem with binary (thresholded) measurements**, i.e. a *set reconstruction problem with a latent 1D manifold*.

Below is a concrete algorithm that actually makes sense computationally, not just notation.

1. What You Are Given

You observe only a binary regime-switch signal:

$$\chi(s) \in \{0, 1\}, \quad s \in \mathcal{S}$$

with

$$\chi(s) = H(\Phi(S(s) - \gamma))$$

Interpreted geometrically:

- There is a hidden curve $\gamma(t) \subset \mathbb{R}^5$
- A scanning map $S(s)$ probes space
- A scalar field Φ changes sign near interaction with γ
- You only see **horizon crossings**

So your observable is effectively:

“Did the scan ray intersect or cross a tubular neighborhood of γ ?”

2. Key Reformulation (Tomography Step)

We convert the binary signal into **constraint surfaces**.

Define:

$$\Omega = \{x \in \mathbb{R}^5 : \Phi(x) > 0\}$$

Then:

- $\chi(s) = 1 \Rightarrow S(s) \in \Omega$
- $\chi(s) = 0 \Rightarrow S(s) \notin \Omega$

So each scan gives a **half-space constraint in function space of γ** .

3. Discretize the Unknown Curve

Represent the curve as a spline / polyline:

$$\gamma(t) \approx \{g_1, g_2, \dots, g_N\}, \quad g_i \in \mathbb{R}^5$$

with smoothness prior:

$$\mathcal{R}(\gamma) = \sum_{i=2}^{N-1} \|g_{i+1} - 2g_i + g_{i-1}\|^2$$

4. Forward Model (Synthetic Tomography Operator)

Define a “hit function”:

$$h(s, \gamma) = \min_{t \in [0,1]} \|S(s) - \gamma(t)\|$$

Then:

$$\hat{\chi}(s; \gamma) = H(\epsilon - h(s, \gamma))$$

So reconstruction becomes:

Find the curve whose tubular neighborhood explains all threshold flips.

5. Inverse Problem (Core Optimization)

We solve:

$$\min_{\gamma} \sum_{s \in \mathcal{S}} \mathcal{L}(\chi(s), \hat{\chi}(s; \gamma)) + \lambda \mathcal{R}(\gamma)$$

where the binary loss is:

$$\mathcal{L} = \chi \log(\hat{p}) + (1 - \chi) \log(1 - \hat{p})$$

but since we have hard thresholds, we replace the Heaviside function with a smooth approximation:

$$H(x) \approx \sigma(kx)$$

So:

$$\hat{\chi}(s; \gamma) = \sigma(k(\epsilon - h(s, \gamma)))$$

6. Tomographic Reconstruction Algorithm (Practical)

Algorithm: Threshold Tomography for Latent Curve Recovery

Input:

- Scan points $S(s_i)$
- Binary regime signal χ_i

Step 0: Initialize Curve

Random walk or PCA initialization:

$$\gamma^{(0)} = \text{principal curve of } S(s_i) \text{ where } \chi = 1$$

Step 1: Compute Distance Field

For each scan point:

$$d_i^{(k)} = \min_t \|S(s_i) - \gamma^{(k)}(t)\|$$

Step 2: Soft Prediction

$$\hat{\chi}_i^{(k)} = \sigma(k(\epsilon - d_i^{(k)}))$$

Step 3: Gradient Update

Minimize:

$$\mathcal{J} = \sum_i (\chi_i - \hat{\chi}_i^{(k)})^2 + \lambda \sum_i \|\Delta^2 \gamma_i\|^2$$

Gradient descent on control points:

$$g_j \leftarrow g_j - \eta \frac{\partial \mathcal{J}}{\partial g_j}$$

Step 4: Geometric Refinement (Critical Step)

After convergence:

- Extract “hit set”:

$$\mathcal{H} = \{S(s_i) : \chi_i = 1\}$$

- Fit a **principal curve / spline through the dense hit manifold**

Use an expectation-maximization principal curve update:

$$g_j^{(k+1)} = \mathbb{E}[x \mid x \in \mathcal{H}, \text{nearest to } g_j]$$

Step 5: Topology Stabilization

Enforce no self-intersections unless the data supports it. Penalize:

$$\mathcal{T}(\gamma) = \sum_{i \neq j} \exp \left(-\frac{\|g_i - g_j\|^2}{\sigma^2} \right)$$

7. Final Algorithm Summary

You are iterating:

1. Guess curve
2. Compute distance-to-curve from scan space
3. Convert to predicted binary regime
4. Compare with observed regime flips
5. Update curve via gradient + projection
6. Enforce smooth + topological regularity

8. What Makes This “Tomography”

Classical tomography:

integrals of a field along rays

Your case:

binary phase-change induced by proximity to a manifold

So instead of reconstructing density, you reconstruct:

A codimension-4 object (curve in $\mathbb{R}^5$) from thresholded sampling of its tubular neighborhood.

This is closer to:

- Medial axis reconstruction
- 1D manifold learning from set-membership queries
- Level-set inverse problems